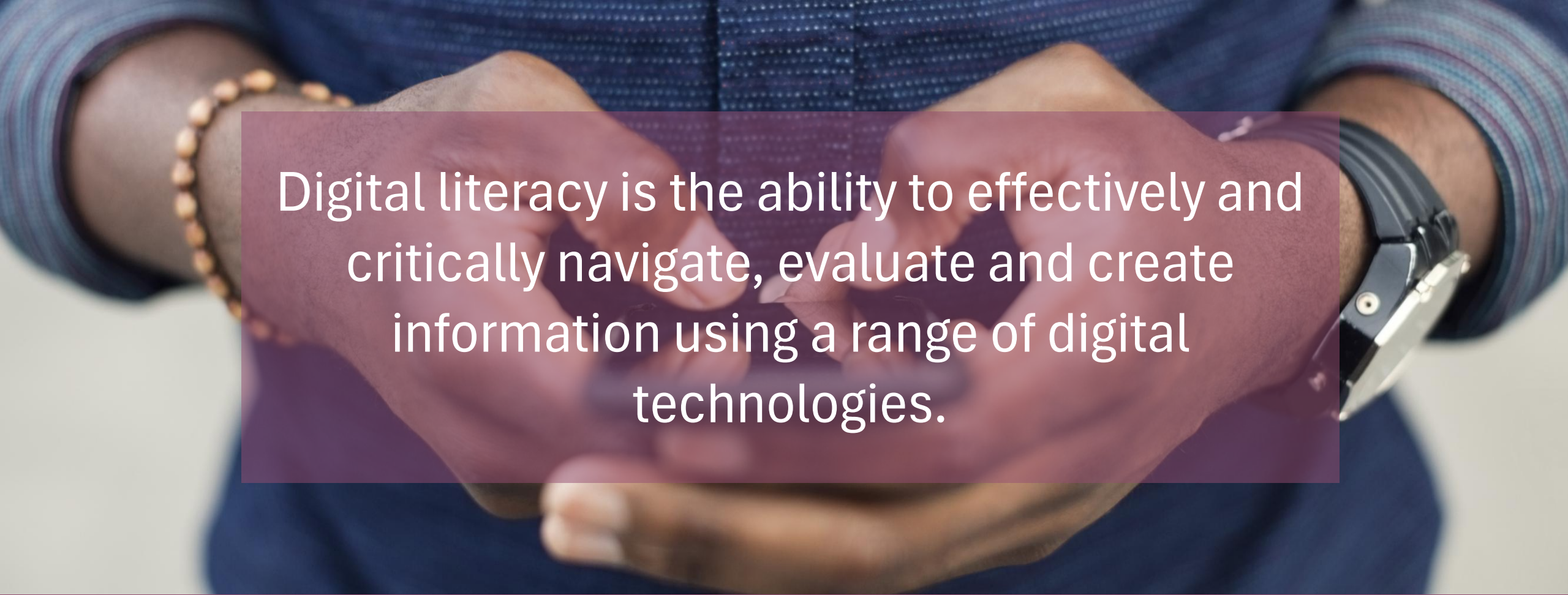




Digital literacy is the ability to effectively and critically navigate, evaluate and create information using a range of digital technologies.

DIGITAL LITERACY

Assoc. Prof. Ts. Dr Roziya Abu | roziya307@uitm.edu.my | +60 104074274

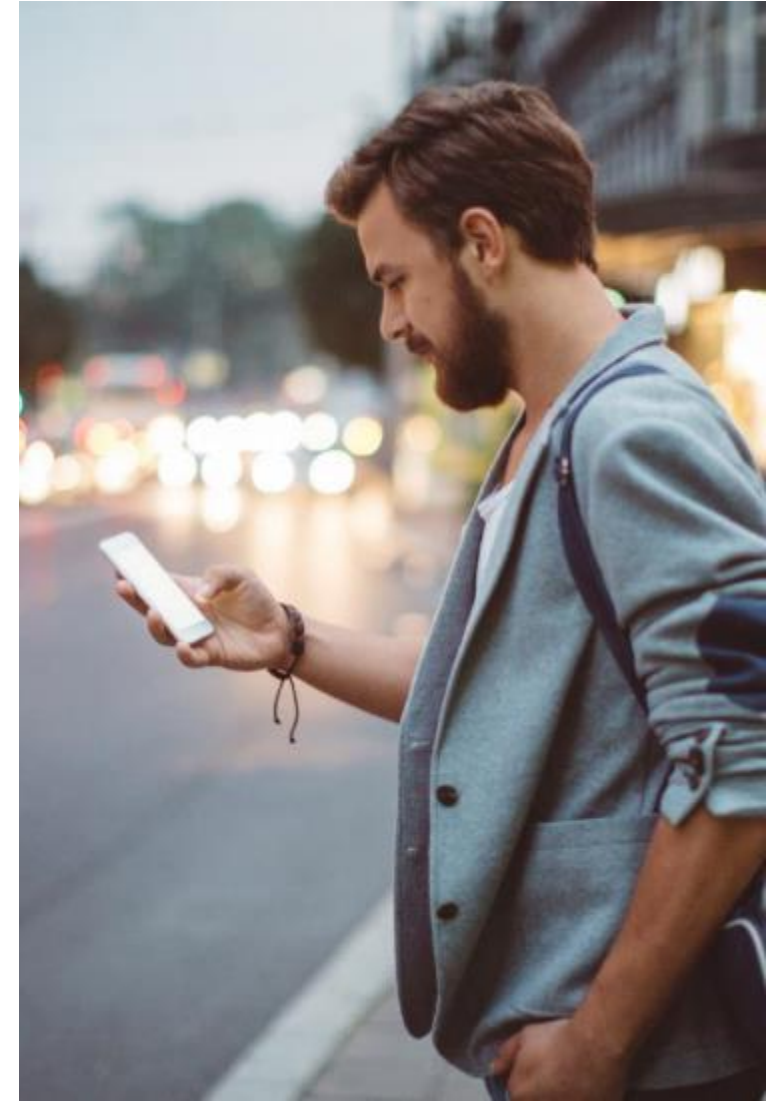
CHAPTER 1

Definition

Models And Theories

Concepts

Evolution



LITERACY



KEMENTERIAN
PENDIDIKAN
MALAYSIA

STATISTIK MEMBACA RAKYAT MALAYSIA

Rakyat Malaysia membaca
15-20 buku secara purata setahun.
Sumber: Adult and Youth Literacy National Regional and Global Trends 1985-2015

Hampir **95 peratus** adalah kadar
celik huruf rakyat Malaysia.
Sumber: Adult and Youth Literacy National Regional and Global Trends 1985-2015

Malaysia diletakkan dalam kedudukan
yang **ke-53** sebagai negara membaca.
Sumber: The World's Most Literate Nation

- Literacy has traditionally been thought of as reading and writing. Although these are essential components of literacy, today our understanding of literacy encompasses much more.
- It refers to the ability to use printed and written information to function in society, to achieve one's goals, and to develop one's knowledge

Malaysian Children & Education

The true shortfall in our education system is staggering—children not attending any school (4.5%) or dropping out of school before reaching Form 5 (18%).

Data as calculated from available sources

Available data on completion rates	SDG 4.1.2 Completion rate (primary, lower and upper secondary education)	
		2022
	Primary education	94.9%* (all children, including those who never attended school)
	Lower secondary education	
	Upper secondary education	~82% (MOE schools only)

Source: Ministry of Education - Quick Facts 2022 and Ministry of Health - Health Indicators MOH 2022

*Data suggests 4.5% of children do not attend school. In addition, 0.6% drop out of MOE schools before completing Standard 6. Hence, Primary Education completion rate = $100 - 4.5 - 0.6 = 94.9\%$.

Data made available by DOSM

SDG 4.1.2 Completion rate (primary education, lower secondary education, upper secondary education)			
 Completion rate increased		2020	2021
	Primary education	98.4	99.0
	Lower secondary education	97.8	99.8
	Upper secondary education	92.8	97.8

Source: Ministry of Education

Source: Sustainable Development Goals (SDG) Indicators, Malaysia, 2021
Department of statistics Malaysia (DOSM)



**Current Malaysia
population
34,809,175**

(Countrymeters.info, April 09, 2025)

23,112,243

Malaysian Adults

ILLITERATE

- or 94.71% of adult population (aged 15 years and above) in Malaysia are able to read and write.

- 1,290,432
Malaysian adults



Literacy is crucial for individual success, societal progress, and economic growth, enabling access to education, employment, and informed decision-making.



IMPORTANT NOTE!

DIGITAL LITERACY DOES NOT REPLACE TRADITIONAL LITERACY but builds onto the foundation.



It is important to remember that digital learning incorporates not only embracing the use of digital resources but being able to critically evaluate the information being presented and distinguish a useful sharing information.



DIGITAL LITERACY CONSISTS OF **EQUIPPING PEOPLE WITH ICT CONCEPTS, METHODS AND SKILLS** TO ENABLE THEM TO USE AND EXPLOIT ICTS.

What is digital literacy?



Digital literacy is the ability to access, manage, understand, integrate, communicate, evaluate and create information safely and appropriately through digital technologies for employment, decent jobs and entrepreneurship. It includes competences that are variously referred to as computer literacy, ICT literacy, information literacy and media literacy (ITU, 2010).

What is digital literacy?

The related concept of information literacy consists of providing people with concepts and training in order to process data and transform them into information, knowledge and decisions. It includes methods to search and evaluate information, elements of information culture and its ethical aspects, as well as methodological and ethical aspects for communication in the digital world

(ITU,2010)

Digital Literacy should be understood....

to mean as the basic skill or ability to use a computer confidently, safely and effectively, including: the ability to use office software such as word processors, email and presentation software, the ability to create and edit images, audio and video, and the ability to use a web browser and internet search engines

(Royal Society, 2012)





Know more...

Definition

- BBC_ What is Digital Literacy :
https://www.youtube.com/watch?v=_LElWqXi7Ag&list=PLcetZ6gSk9682A7ZAZq2s9lqB-y8Ng63e
- What is Digital Literacy?
Cultivating Digital Literacy:
<https://www.youtube.com/watch?v=oU1X3QpX-90>
- Digital literacy :
<https://www.youtube.com/watch?v=eRBJORugRNU>





DIGITAL LITERACY MODELS AND THEORIES

Digital Literacy: MODELS AND THEORIES

Consist of conceptual frameworks and academic theories used to explain how individuals acquire, develop, and apply digital literacy skills in various contexts, including education, the workplace, and daily life.



Digital Literacy Model 1:
Multiple Literacies Model (*Multiliteracies*)



Digital Literacy Model 2:
Digital Literacy Framework Spires & Barlett



Digital Literacy Model 3:
Steve Wheeler's Digital Literacies



Digital Literacy Model 4:
JISC Digital Capability Framework

Digital Literacy: MODELS AND THEORIES

Multiple Literacies Model (Multiliteracies)

a set of interconnected literacies:

1

Multiple Literacies Model



Core literacies introduced by the New London Group (1996)

ujs.ca/octac

Multiple Literacies Model...

- The Multiple Literacies Model (sometimes called Multiliteracies) was first introduced by the New London Group in 1996.
- Their seminar paper, *A Pedagogy of Multiliteracies: Designing Social Futures*, responded to two major global shifts:
 - Cultural and linguistic diversity due to globalization.
 - Technological transformation in communication and media.

1. Digital Literacy
2. Media Literacy
3. Information Literacy
4. Visual Literacy
5. Critical Literacy
6. Cultural Literacy
7. Health Literacy, Financial Literacy, Environmental Literacy, etc. – context-specific literacies.

Multiple Literacies Model (Multiliteracies)

1. Digital Literacy – navigating, evaluating, and creating information using digital technologies.
2. Media Literacy – critically analyzing media messages and understanding media's role in society.
3. Information Literacy – locating, assessing, and using information effectively.
4. Visual Literacy – interpreting and creating meaning from images, videos, and visual formats.
5. Critical Literacy – questioning power structures and ideologies behind texts and media.
6. Cultural Literacy – understanding and respecting diverse cultures and perspectives.
7. Health Literacy, Financial Literacy, Environmental Literacy, etc. – context-specific literacies.

This model acknowledges that literacy is contextual, socially situated, and multimodal—meaning it involves reading and producing content across different formats (text, video, image, sound, etc.).

Digital Literacy

Multiple Literacies Model (Multiliteracies)

1. Digital literacy is a key aspect of multiliteracies, emphasizing the skills needed to navigate and engage with the digital world.
2. Within the Multiliteracies model, digital literacy explains the ability to effectively use, interpret, and create meaning across various digital media and technologies, encompassing skills like finding, evaluating, and communicating information online.



Media Literacy

Multiple Literacies Model (Multiliteracies)

1. ability to analyze, evaluate, and create messages in a wide variety of media modes, genres, and formats – texting, blogging, gaming, writing for a website, participating in a forum, creating a video, a podcast, interpreting complex messages from a wide range of multimedia and transmedia.
2. Media literate people can skillfully create and produce media messages, both to show understanding of the specific qualities of each medium, as well as to create media and participate as active citizens.

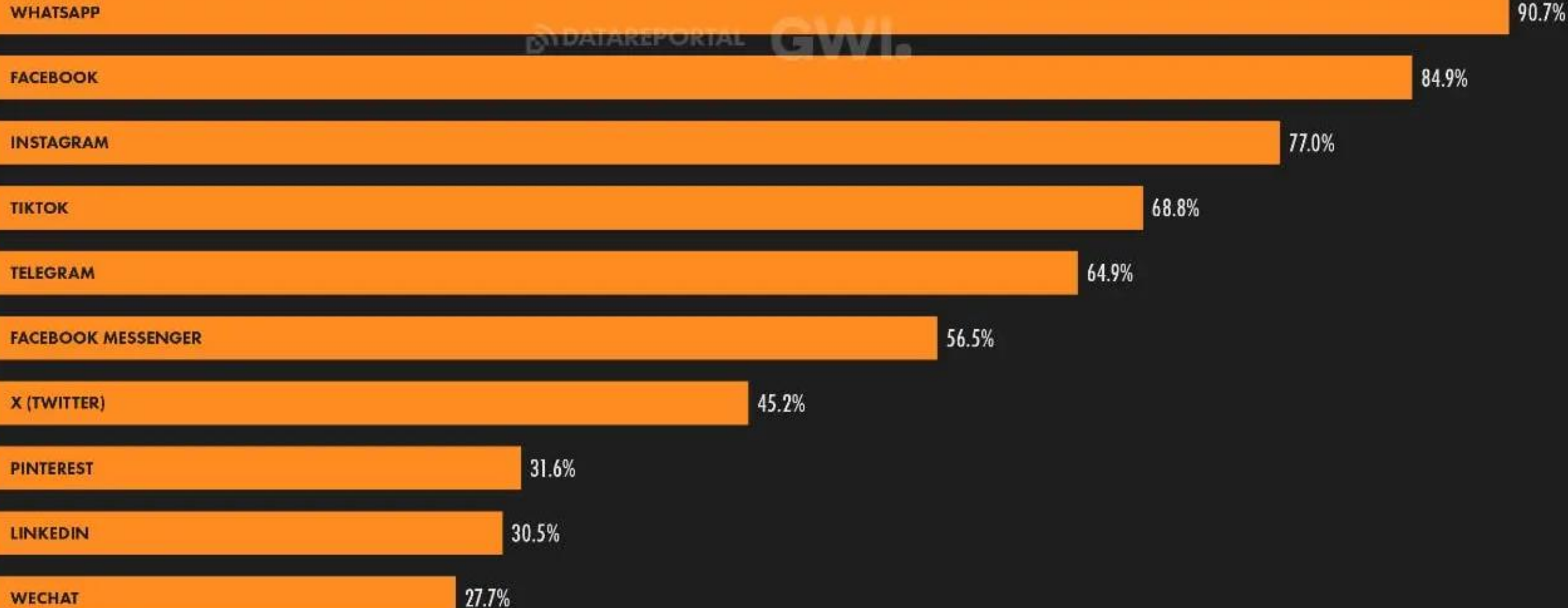


JAN
2024

MOST USED SOCIAL MEDIA PLATFORMS

PERCENTAGE OF INTERNET USERS AGED 16 TO 64 WHO USE EACH PLATFORM EACH MONTH

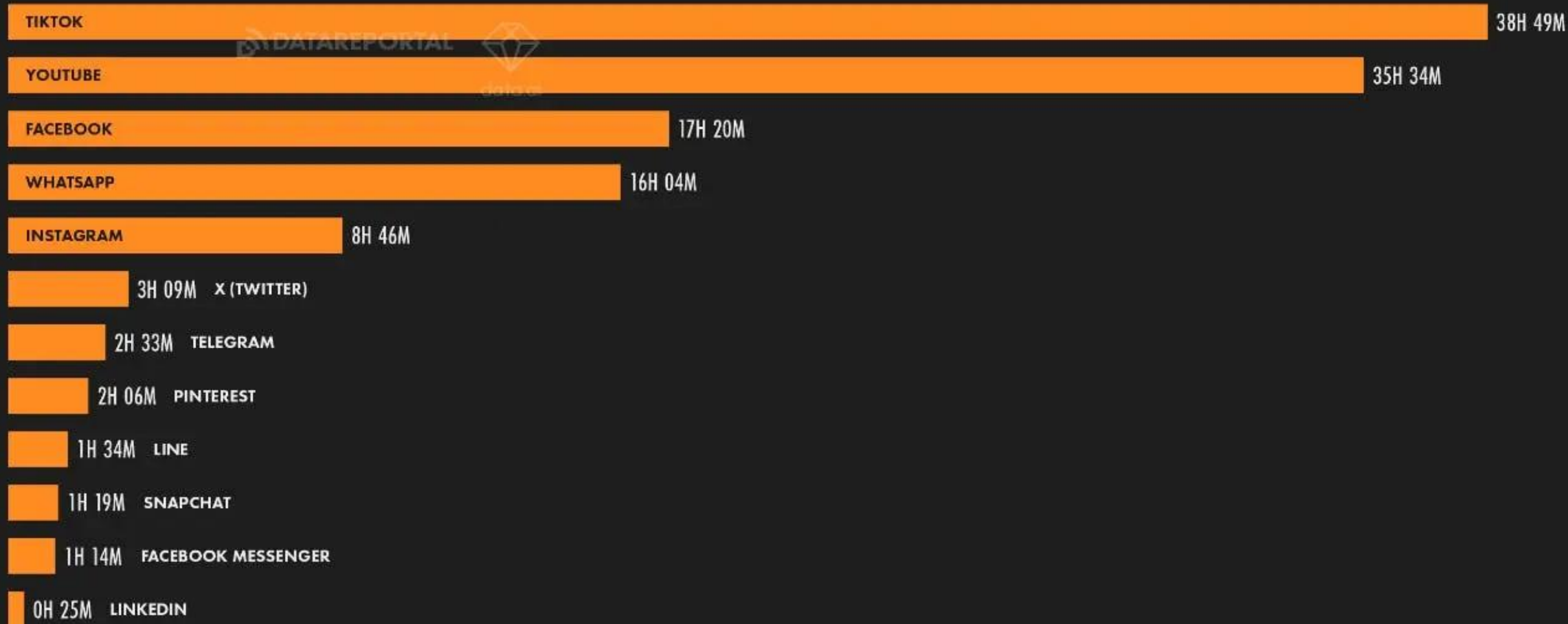
NOTE: YOUTUBE IS NOT OFFERED AS AN ANSWER OPTION FOR THIS QUESTION IN GWI'S SURVEY, SO IT WILL NOT APPEAR IN THIS RANKING.



JAN
2024

TIME SPENT USING SOCIAL MEDIA APPS

AVERAGE TIME PER MONTH THAT ACTIVE USERS SPENT USING EACH PLATFORM'S ANDROID APP BETWEEN 01 JULY AND 30 SEPTEMBER 2023



Information Literacy

Multiple Literacies Model (Multiliteracies)

1. the ability to access, critically evaluate, understand, store, remix and apply information –for the purpose.
2. As we deal with so much information every day, we need to be able to effectively manipulate it to meet our needs.
3. It means knowing how to find, evaluate, and use information effectively and ethically.



Visual Literacy

Multiple Literacies Model (Multiliteracies)

1. the ability to interpret, negotiate, and make meaning from information presented in the form of an image: understanding how visual elements such as line, colour, shape, texture, space, symbols, pattern and composition create meaning.
2. The internet is a visual medium, and important information is not always conveyed in written, linear text format as it used to be.



The Importance of Visual Literacy



image credit: Veer edutopia.org/blog/ccia-10-visual-literacy-strategies-todd-finley

<https://www.youtube.com/watch?v=svtaCLMACGg&t=32s>

Critical Literacy

Multiple Literacies Model (Multiliteracies)

1. the ability to evaluate a text (visual, aural, written, spoken) and see beyond the explicit message to understand the implicit messages including bias, political or social meanings.
2. Its important now because information online goes through no editing process, and it is up to the person using the information to discern if it is accurate, unbiased and appropriate

how to decode, make meaning, use, and critically analyze multiple text types (including those beyond traditional print) for diverse purposes and in various contexts, recognizing the power dynamics and social contexts embedded within those texts.



Stop



Stop, children crossing



Give way



No entry



No lorry



No Motorcycle



No slow moving vehicles



No U-Turn



No left turn



No right turn



No overtaking



No overtaking for heavy vehicles



Heavy vehicles prohibited on the fast lane



No hawking sign



No parking



No stopping



No stopping, except emergency sign



No honking



Width clearance 2.5 metres



Height clearance 5 metres



Height limit sign at gantry

Cultural Literacy

Multiple Literacies Model (Multiliteracies)

1. cultural literacy explains the importance of understanding and appreciating cultural differences and diverse perspectives within communication and meaning-making, recognizing that literacy practices are shaped by cultural, social, and political contexts.
2. Cultural literacy is not just about decoding and encoding text, but also about understanding how cultural contexts, social structures, and power dynamics shape meaning-making.





Dr. Nicole A. Cooke

Illustrated by Alejandra Diaz

Digital Literacy: MODELS AND THEORIES

Digital Literacy Framework Spires & Barlett

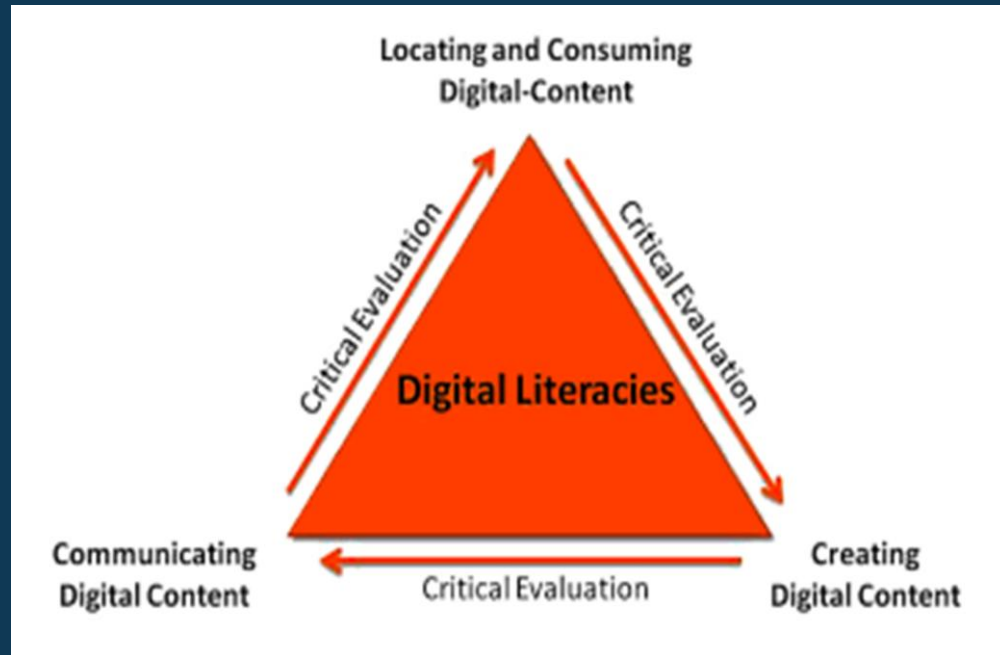
emphasize cognitive engagement, ethical participation, and the capacity to use digital tools effectively for meaning-making and communication.

2

Digital Literacy Framework

by Hiller A. Spires and Monica A. Bartlett (2012)

Spires and Bartlett defines digital literacy as more than just technical skills.



Three core components

1. Finding and critically evaluating information
2. Creating digital content
3. Communicating and collaborating with others

1. Digital Literacy Framework Spire & Barlett

Finding and Critically Evaluating Information



beyond simple search skills



Involves **information literacy, credibility assessment, bias detection**, and the ability to **distinguish between reliable vs. unreliable digital sources.**



Encourages critical thinking in navigating the **overload of online information.**

2. Digital Literacy Framework Spires & Barlett

Creating Digital Content



Digital literacy involves **production**, not just consumption



Includes **multimedia composition**, hypertext creation, social media use, blogs, wikis, etc..



Stresses the importance of **digital authorship** and **visual/media communication skills**.

3. Digital Literacy Framework Spire & Barlett

Communicating and Collaborating with Others



Encompasses **online interaction**, **digital collaboration tools**, and **networked communication**.



Emphasizes **digital citizenship**, including **netiquette**, **ethical behavior**, and **respectful participation** in digital communities.

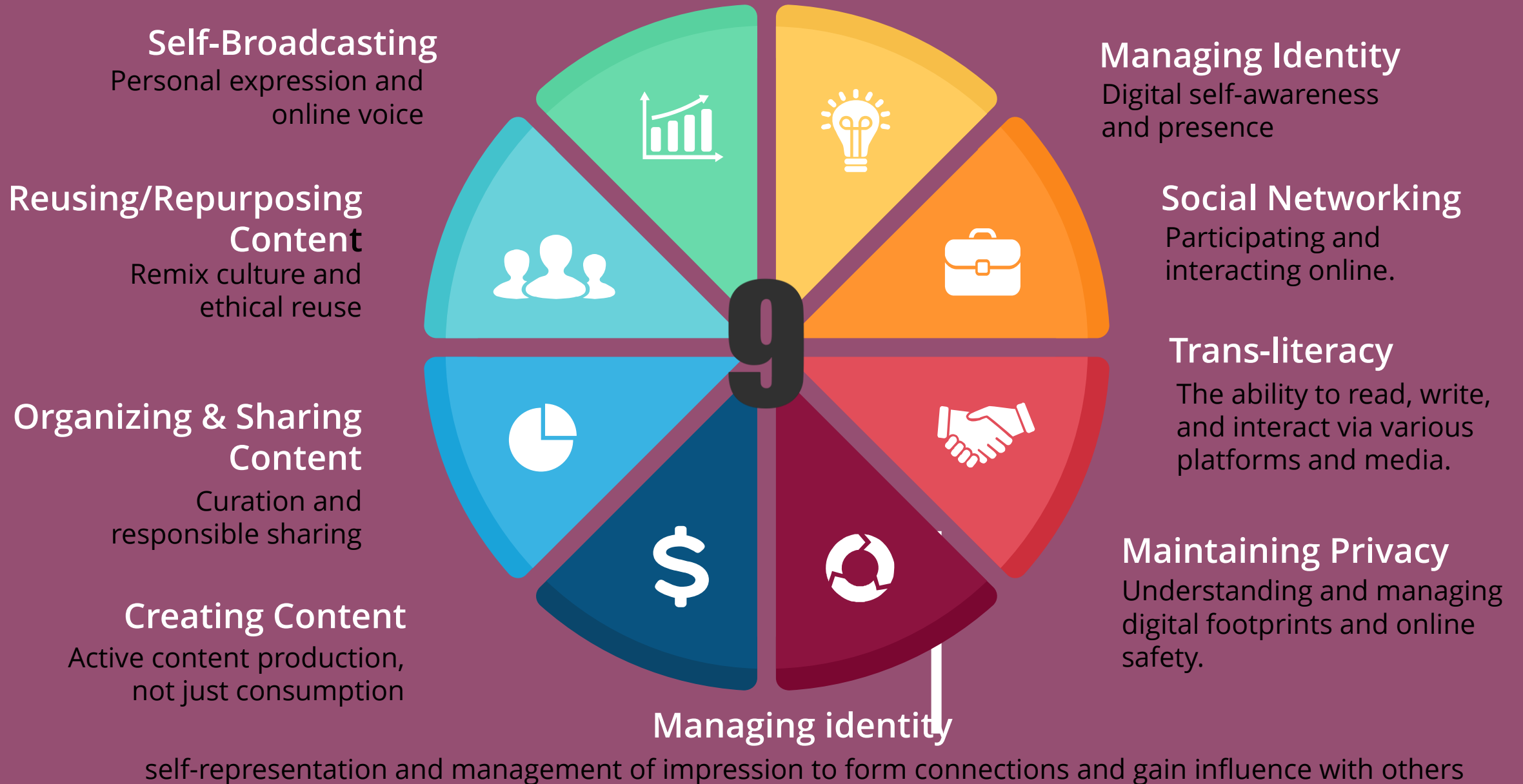
Digital Literacy: MODELS AND THEORIES

Steve Wheeler's Digital Literacies

emphasis on social, participatory, and evolving aspects of learning in the digital age.



Steve Wheeler Digital Literacies





Court postpones verdict in Aliff Syukri obscene content case

By **Bernama** - March 11, 2025 @ 2:25pm



- <https://youtu.be/FBTnJzIVakM>

MALAYSIA

‘Influencer’ Adeline Chang fined RM7,000 over offensive remarks on minor’s pic

- https://www.youtube.com/shorts/-54fISD_94U





MCMC to fine Harith Iskander, Cecelia Yap RM10,000 each

FMT Reporters - 05 Mar 2025, 08:06 PM

- <https://youtu.be/-UGD8j5L2cl>

Lorry driver in influencer Esha's cyberbullying case gets 12 months prison

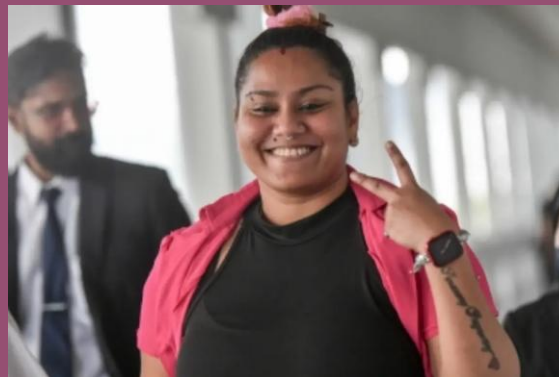
He was charged with knowingly creating and posting obscene content through a communication service with the intent to offend others via the TikTok account "@dulal_brothers_360" at 10.12 pm on June 30, which was later viewed at 10.15 pm on June 30 at Jalan Ipoh, Sentul.

He was charged under Section 233(1)(a) of the Communications and Multimedia Act 1998, which carries a maximum fine of RM50,000 or imprisonment of up to one year, or both, with an additional fine of RM1,000 imposed for each day the offense continues after conviction.

Sathiskumar also faced a separate charge for insulting and outraging the modesty of Rajeswary's mother, P.R. Puspa, 56, through the same TikTok profile on the same date, time, and location.

Woman fined for insulting influencer now claims she is cyberbullying victim

By Luqman Hakim, Amisha Syahira - January 10, 2025 @ 7:54pm



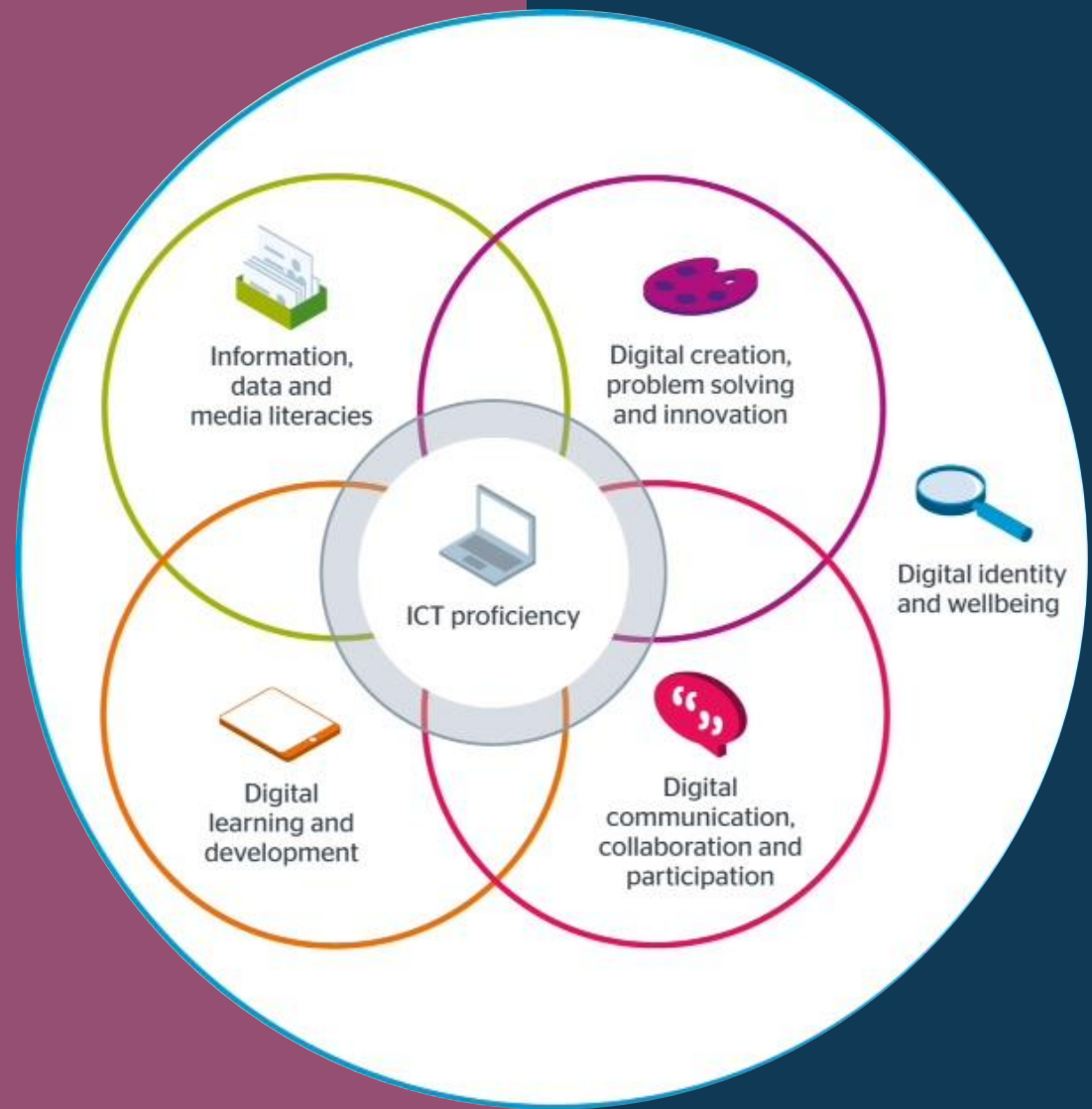
Digital Literacy: MODELS AND THEORIES

JISC Digital Capability Framework

4

JISC Digital Capability Framework

JISC (Joint Information Systems Committee) is a UK-based organization that supports digital transformation in education and research.



- Originally launched in 2015, and regularly updated (latest version 2022), the framework outlines the digital knowledge, skills, and attitudes needed for effective participation in a digital society and economy.

JISC digital capability six core elements - each representing a domain of competence:



1. DIGITAL IDENTITY AND WELLBEING

Managing digital reputation and presence.

Ensuring personal digital safety and work-life balance.

Understanding digital rights and responsibilities.

2. INFORMATION, DATA, AND MEDIA LITERACIES

The ability to critically find, evaluate, manage, and communicate information in various forms.

Data literacy includes interpreting and using data effectively.

Media literacy involves interpreting media messages and creating content.

JISC digital capability six core elements - each representing a domain of competence:

3. DIGITAL COMMUNICATION, COLLABORATION, AND PARTICIPATION

Engaging in digital communication via emails, chats, forums, and social media.

Collaborating using digital tools and participating in online communities.



4. DIGITAL LEARNING AND DEVELOPMENT

Using digital tools for personal and professional learning.

Reflecting on digital practices and identifying learning needs.

JISC digital capability six core elements - each representing a domain of competence:

5. DIGITAL CREATION, PROBLEM SOLVING, AND INNOVATION

Creating digital content (text, images, videos, data visualizations).

Adopting digital solutions for complex tasks or problems.

Innovating with digital tools.



6. ICT PROFICIENCY (FOUNDATION)

Functional IT skills: using software, devices, and operating systems. Includes productivity software, basic troubleshooting, and efficient use of hardware.



[https://www.youtube.com/shorts](https://www.youtube.com/shorts/DkCnu3ccBGI)
[/DkCnu3ccBGI](https://www.youtube.com/shorts/DkCnu3ccBGI)

CONCEPTS

OF DIGITAL LITERACY

Digital literacy means having the skills you need to live, learn, and work in a society where communication and access to information is increasingly through digital technologies like internet platforms, social media, and mobile devices.

BROADER PERSPECTIVES

UNESCO links digital literacy to lifelong learning and sustainable development, seeing it as critical for education, employment, and active citizenship.

It is also closely tied to media literacy, information literacy, and ICT literacy.

CORE COMPONENTS OF THE CONCEPTS OF **DIGITAL LITERACY**

The ability to effectively and critically navigate, evaluate, create, and communicate information using a range of digital technologies



ACCESSING AND
EVALUATING INFORMATION



CREATING AND
COMMUNICATING
CONTENT



DIGITAL COMMUNICATION
AND COLLABORATION



CRITICAL THINKING
AND PROBLEM
SOLVING



UNDERSTANDING
DIGITAL RIGHTS
AND RESPONSIBILITIES



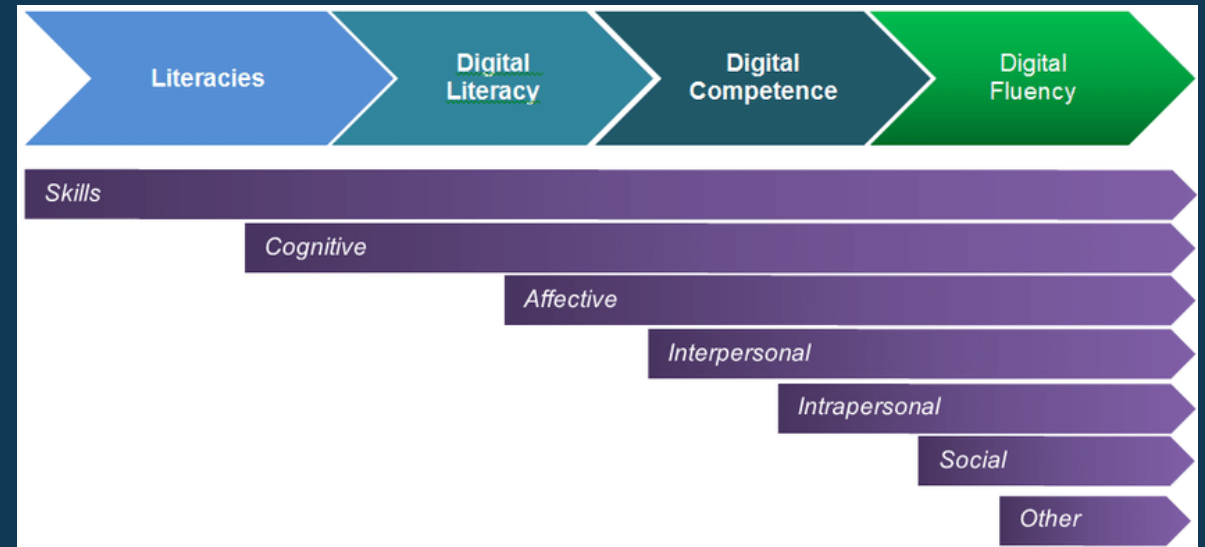
DIGITAL IDENTITY
AND FOOTPRINT
MANAGEMENT



CYBERSECURITY
AWARENESS

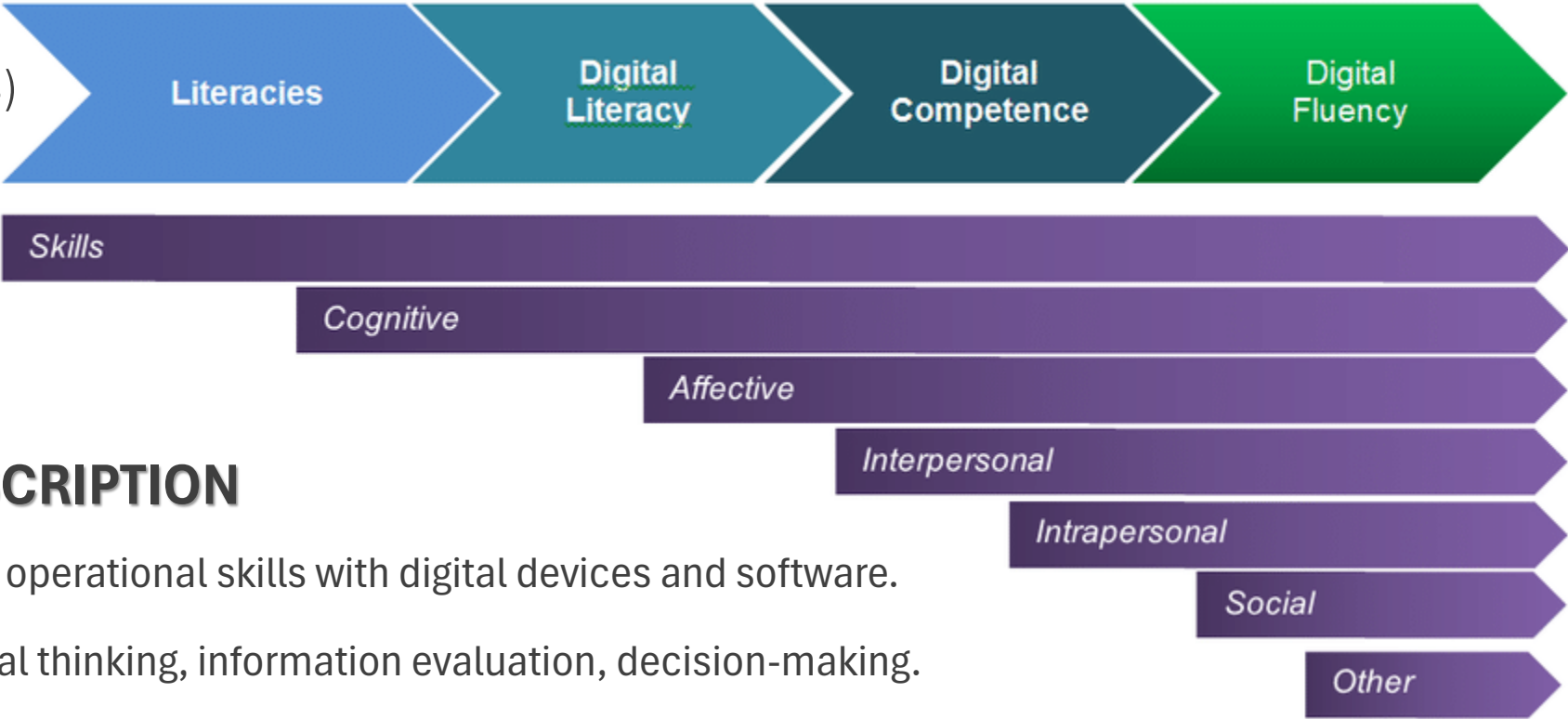
DIGITAL LITERACY EVOLUTION

Erhan Sinay (2018), conceptualized the evolution of digital literacy as a **progressive journey**—moving from basic technical skills to more critical, ethical, and civic-oriented uses of digital tools and environments.



Erhan Sinay (2018)

Erhan Sinay (2018)



LAYER

 Skills

 Cognitive

 Affective

 Interpersonal

 Intrapersonal

 Social

 Other

DESCRIPTION

Basic operational skills with digital devices and software.

Critical thinking, information evaluation, decision-making.

Emotional responses and attitudes towards technology (e.g., motivation, confidence).

Collaborative skills and communication in digital contexts.

Self-reflection, self-regulation, and managing one’s own digital presence.

Understanding digital citizenship, social norms, and online behavior.

Includes ethical reasoning, cultural sensitivity, and adaptability in digital spaces.

Digital Literacy: DEFINITIONS

Further discussion...



Digital literacies include data literacy, information literacy, visual literacy, media literacy, and metaliterary, as well as related capacities for assessing social and ethical issues in our digital world

(ELI7169, 2019)

Digital Literacy: DEFINITIONS

Further discussion.....



Digital literacy is the ability to understand and use information in multiple formats from a wide variety of sources when it is presented via computers and the internet.

(Gilster, 1997)

Link:<https://slideplayer.com/slide/5761794/>

Digital Literacy: DEFINITIONS



Digital Literacy defined by Five Subset Categories: (Eshet-Alkalai, 2004) :

Photo-visual literacy	ability to read and deduce information from visuals
Reproduction literacy	ability to use digital technology to create a new piece of work or combine existing pieces of work together to make it your own
Branching literacy	ability to successfully navigate in the non-linear medium of digital space.
Information literacy	ability to search, locate, assess & critically evaluate information found on the web.
Socioemotional literacy	social and emotional aspects of being present online, whether it may be through socializing, and collaborating, or simply consuming content.

AREA OF DIGITAL LITERACY



NO.	AREA	DESCRIPTION
1	Information Literacy	Ability to find, evaluate, and use information effectively and ethically across digital platforms.
2	Media Literacy	Understanding, analyzing, and creating messages across various media formats.
3	ICT Literacy	Proficiency in using digital devices, communication tools, and networks to access and manage information.
4	Digital Communication	Using digital technologies to communicate, collaborate, and share information with others.
5	Digital Etiquette	Understanding proper and responsible behavior when using digital devices and interacting online.
6	Digital Law	Awareness of legal rights and responsibilities related to digital content, including copyright and privacy.

AREA OF DIGITAL LITERACY

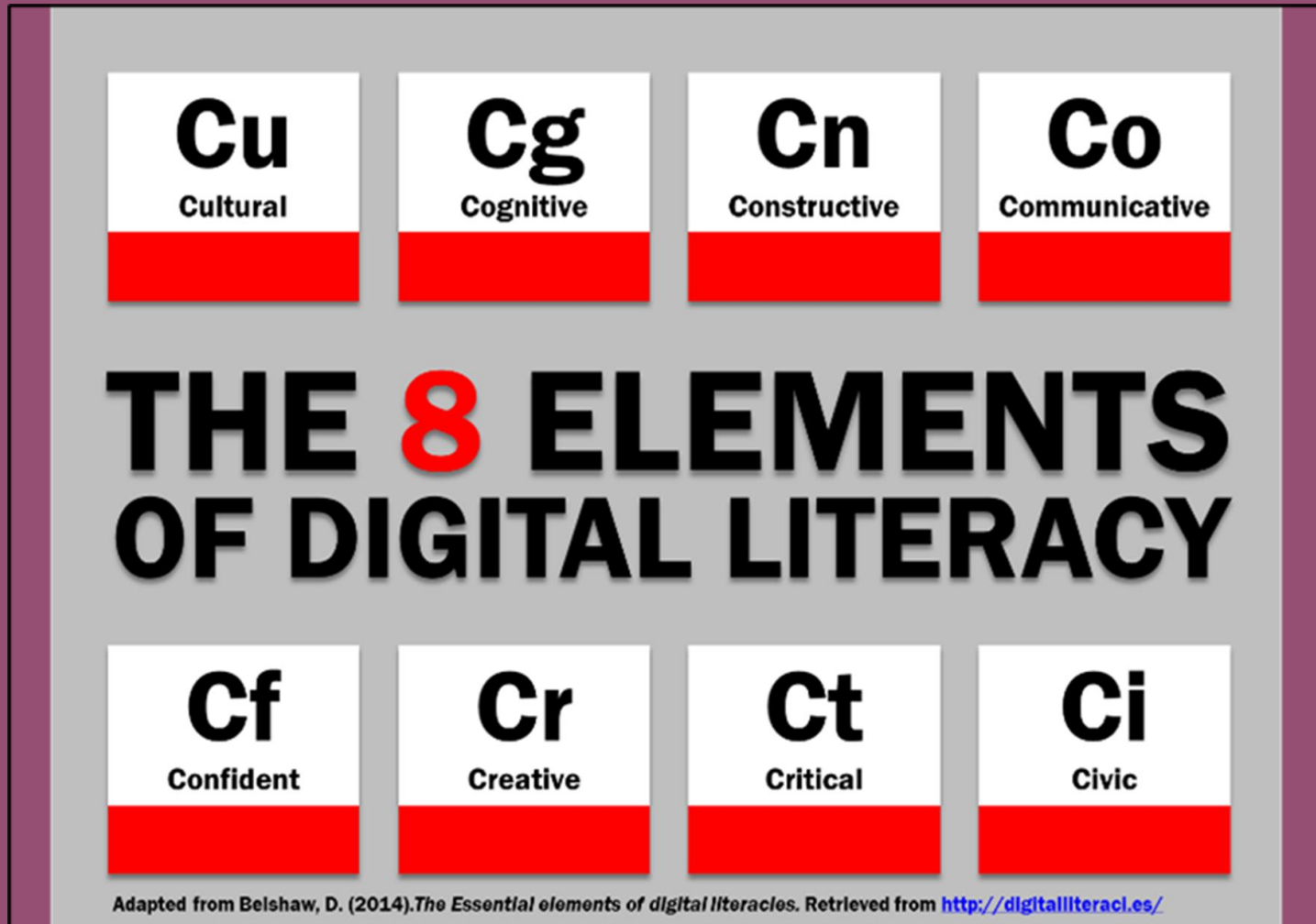


NO.	AREA	DESCRIPTION
7	Digital Rights & Responsibilities	Knowing and respecting one's digital rights and upholding responsibilities in online spaces.
8	Digital Health & Wellness	Maintaining physical and psychological well-being in a digital world, including screen time management.
9	Digital Security	Protecting digital identity and information through secure behavior (e.g., passwords, privacy settings).
10	Digital Access	Advocating for and ensuring equal access to technology and internet for all individuals.
11	Digital Commerce	Understanding and responsibly participating in the digital economy, including e-commerce and online transactions.
12	Critical Thinking & Problem Solving	Applying analytical skills to navigate, assess, and solve challenges in digital environments.

The eight essential digital competencies (8Cs):

Cultural,
Cognitive,
Collaborative,
Communicative,
Confident,
Creative,
Critical, and
Civic

Belsaw (2018).



PRINCIPLES OF DIGITAL LITERACY

COMPREHENSION

reach the level of comprehension of digital literacy through lessons, home practice and support group work. Grasp basic concepts of safety and digital literacy.



SOCIAL FACTORS

there are social influences and factors in having an online digital presence



INTERDEPENDENCE

different digital platforms are dependent on each other. media forms not simply co-exist, but supplement one another



CURATION

curating own content and digital information. using platforms that offer the ability to store and save content for later



**DIGITAL
TRANSFORMATION
IS NOT ABOUT
TECHNOLOGY
AT ALL.
ITS ABOUT **PEOPLE****

